

Design for Additive Manufacturing

Redesign of aircraft driver system

by

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Introduction

An aircraft company wants to redesign an aircraft driver system and consolidate its structural components. This assignment discusses the exploration process of designing a replacement driver using topology optimisation, generative design and additive manufacturing techniques. The old driver design is depicted in Figure 1 and has a weight of 361 grams.

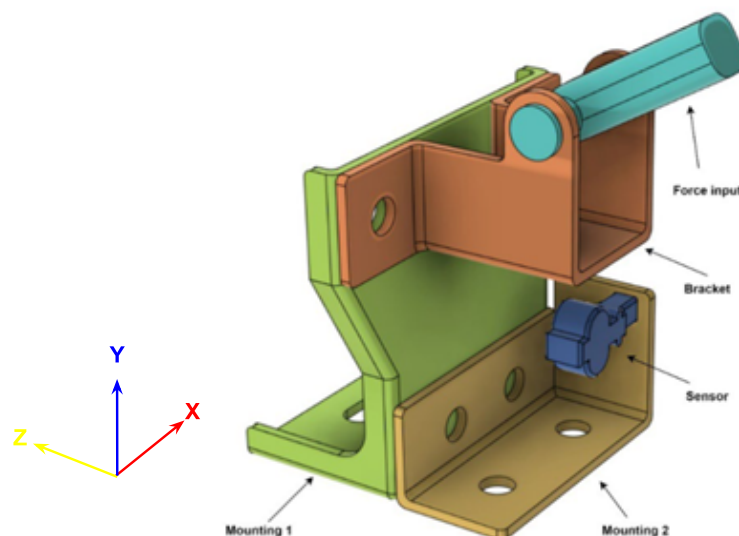


Figure 1: Current aircraft driver.

This assignment aims to redesign the aircraft driver, with a minimal weight reduction of 50%, and maintain structural integrity while being the most cost-efficient driver possible. In addition, the secondary aim is to consolidate the three structural parts into one, not counting the sensor and force input handle. The forces on the handle sensor are given in Table 1. In this assignment, Autodesk Fusion 360 is used for the topology optimisation process, while SimuFact Additive is used for printing simulation and optimisation of the part.

Table 1: Loads on the driver

Loads	Locations [x,y,z] (mm, from origin)	Magnitude (N)
Force 1 (Z-direction)	[10.00, 113.00, -38.828]	1500
Force 2 (Y-direction)	[10.00, 133.00, -38.828]	-1000
Force 3 (Z-direction)	[-20.00, 37.00, -28.28]	-1000

Table 1: Forces acting on the driver.

Applying these forces on the original driver results in high deformation and a low safety factor (SF) around the attachment point of the sensor, and a high SF in the rest of the model with a high deflection of 3.01 mm at the top attachment points (Figure 2 & 3). An additive manufacturing part not only would reduce the mass, but also improve load distribution and therefore stress and consequent displacement.

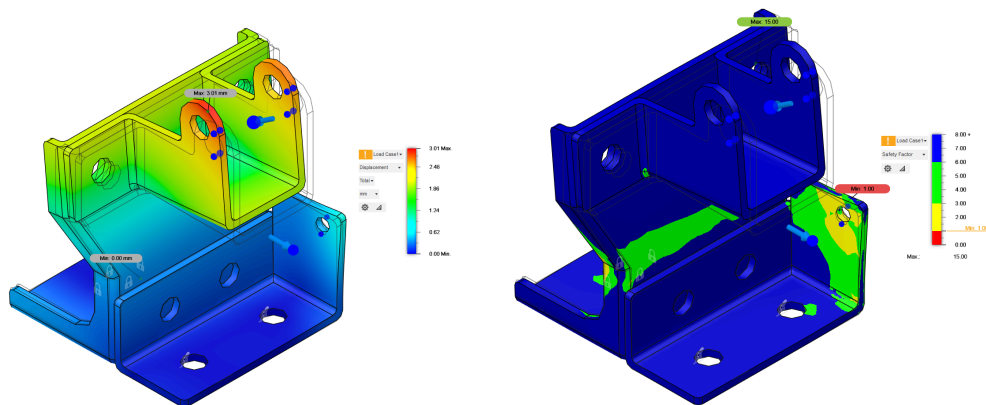


Figure 2 & 3: Deformation of original driver (left) and Safety Factor (right).

Strategy Topology Optimisation

The introduction gave a quick overview of some of the requirements for the driver. However, those are not all of the requirements for the design. Below is a list with all the requirements stated to understand how to design the driver with the final goal of cost efficiency:

Requirements:

1. The driver must consist of a single structural component.
2. The driver must be manufactured using Laser Powder Bed Fusion.
3. The driver must have a minimum safety factor of at least 1.5, conforming to the aircraft standards (*Federal Aviation Administration*, 2023).
4. The driver must have a weight reduction of at least 50%.
5. The mountings must be reachable for installation and maintenance of the driver.
6. The sensor must be reachable for installation and maintenance.

1.1 Initial Topology Optimisation

The initial design space for topology optimisation was created to be as large as possible so the algorithm could create an optimal shape given the applied forces, assuming the original model's outer edges as the max dimensions. Volume was removed around connection points to ensure every connection remained accessible during installation and maintenance. The initial simulation included pinned constraints at the mounting points as well as a frictionless constraint at the bottom plane; this was later removed as it was determined to provide the wrong anchor points for the simulation. Forces were applied as shown in Table 1, the largest possible design space was given and the material from the old driver system was applied, namely Titanium 6Al-4V. This resulted in the geometry as seen in Figure 4, these results show the importance of the area around the sensor mounting and illustrate that the pillars leading to the input force don't require bridging.

Results given by Fusion showed to be bulky and fail to reduce weight compared to the initial design. It is assumed that this is caused by the method by which Fusion generates the Topology optimisation, which differs from the functioning in Siemens NX and SolidWorks. The illustration in Figure 4 shows the first result, and Figure 5 shows the final iteration where the design space was optimised by removing design space and providing better clearance around the accessibility locations. Attempts were made to lower the resulting weight in certain iterations by setting the target mass at a lower percentage. While the mass goal was achieved at a volume target of approximately 5%, going under 20% resulted in disconnecting geometry.

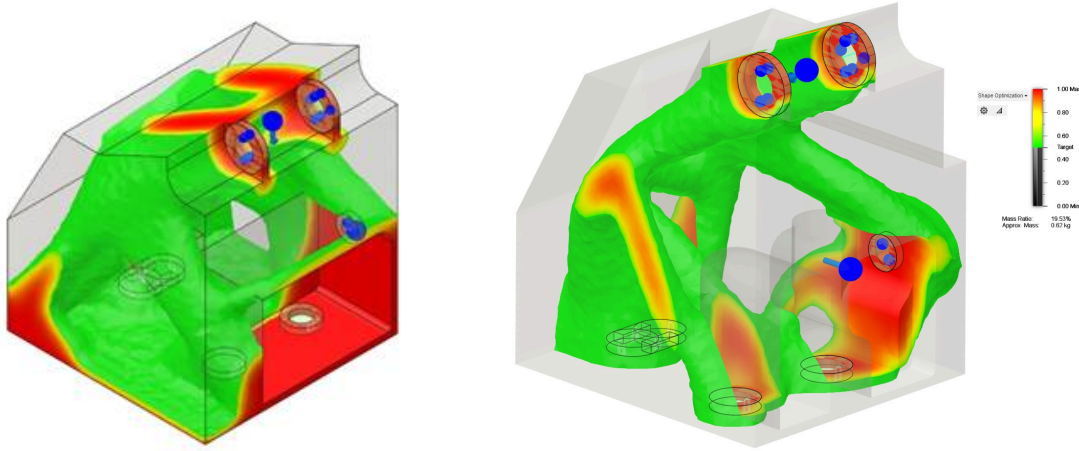


Figure 4 & 5: First and last topology optimisation

1.1.1 Problems with Topology Optimisation

During the iterations of topology optimisations, it became clear that the algorithm favoured geometry which contributes more to the stiffness of the part, resulting in parts disconnecting while keeping a lot of material around connection points. This methodology would not create an optimisation that would fulfil the weight requirements.

Therefore, it has been concluded that Fusion's topology optimisation function is unsuitable for use with a larger design space and limited constraints, as it will result in bulky designs. A new approach was required to optimize the results. Three additional approaches were taken, namely:

1. A generative design approach;
2. A generative design, followed by topology optimisation using the resulting geometry;
3. A manually created design based on findings from the topology optimisation.

The following sections explain the strategies of these approaches and discuss their results.

1.2 Generative Design

The first alternative step taken to create other possible solutions was creating a generative design solution; this was also done in Autodesk Fusion 360. This design also needs to meet the requirements as stated in Section 1. Therefore, obstacle geometry was placed at the places on the driver where maintenance and installation are required. In addition, the locations of the fasteners and force hinge are the preserved geometry. This can all be seen in Figure 6 below.

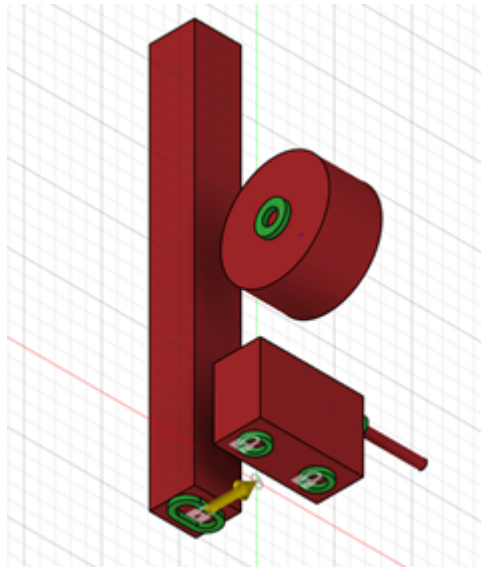


Figure 6: Study generative design geometries

Several simulations were done using generative design, using different studies. The first simulation used a study with a minimum safety factor of 1.5 with the objective of minimising mass. This yielded the model, which can be seen in Figure 7.



Figure 7: Generative design model with SF 1.5 and its properties

Several things can be learned by analyzing this model. Firstly, the general shape of the model is the two backbones going from the hinge to the larger rear fastener, while also connecting to the two front fasteners, creating a triangular shape. Secondly, all the backbones have a cross connection to a central, more solid part, which functions as a centre for the design.

This centre also connects the forces of the sensor to the backbones, making use of a thick horizontal connection.

The second generative design uses a larger safety factor. This model can be used as a design space for topology optimisation, already making the shape more constricted and suitable for additive manufacturing. The resulting model can be seen in Figure 8.

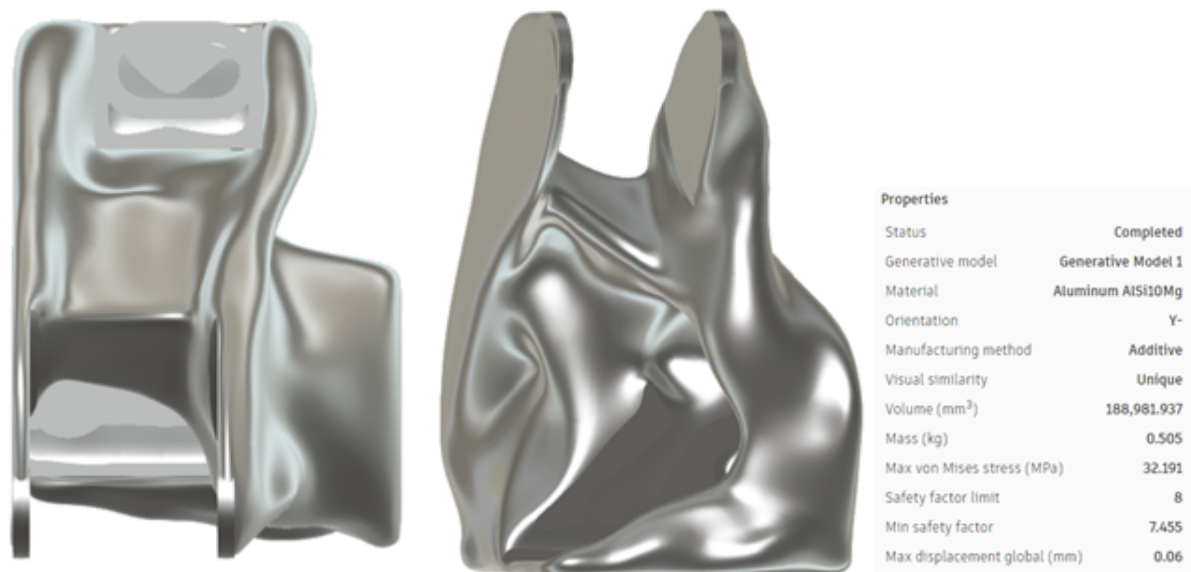


Figure 8: Generative Design model with SF of 7.

1.3 Topology Optimisation on Generative Design

The second approach that has been taken was to use the generative design outcome with a high minimum safety factor mentioned above as a design space for a topology optimisation. The goal was to reduce the weight of the high safety factor generative design model and reduce the part to the effective geometry. The same constraints and loads were applied as in the other optimisations.

The result of the topology optimisation is depicted in Figure 9 below. It weighs 312 grams. Similar to the initial topology optimisations, it was still found that there was a clumping of material. The hard surfaces also indicated that the optimisation could not place material on its optimal spots.

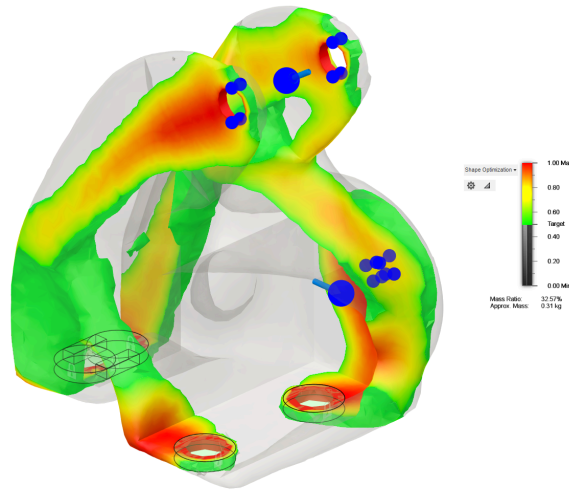


Figure 9: Topology optimisation with a generative design as design space.

1.4 Manual design based on TO result

The final approach to alter the structure was a manual design based on the shape of the last successful topology-optimized part. Although initial topology optimisations done earlier during the project remained too heavy, the results still indicated optimal load paths for the loadcase. The approach and result can be seen in Figures 10 & 11 below. The design was made by creating a smooth skeleton using the Fusion 360 “Forms” function which followed the shape of the topology-optimised part and removing abundant material which the topology optimisation failed to remove. During static simulations, the first iteration showed high safety factors over the model, implying that the 160-gram design could be optimised. Further iterations removed material around high SF areas and reinforced areas where an SF below 1.5 was found. Finally, this methodology created a driver with a weight of 102.83 grams, a minimum SF of 1.69 and a maximum displacement of 0.78 mm under load.

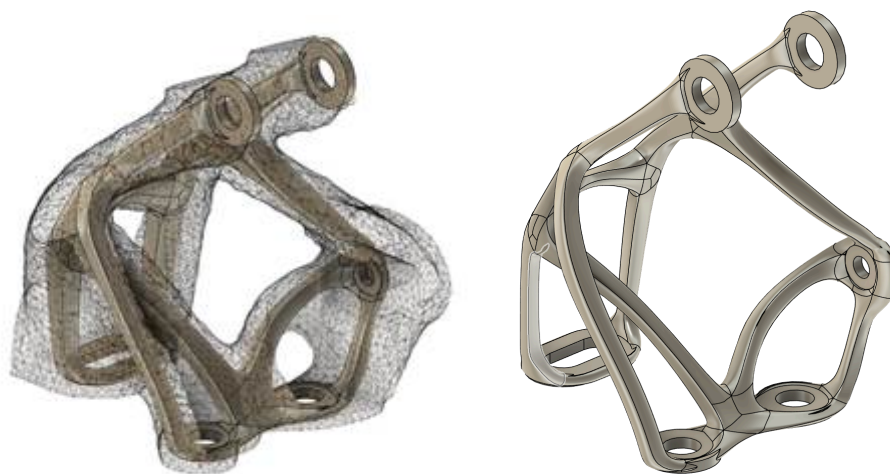


Figure 10 & 11: Skeletal design based on topology optimisation. First and last iteration.

1.5 Comparison of different approaches

While the generative design and manual design show promising (mechanical) properties, the topology-optimised generative design exhibits the same problems as the original topology optimisation. While the design achieved a lower weight of 312 grams, the requirement of halving the original design's weight was not met. It was therefore decided to continue with alternative approaches 1 and 3. As both designs showed similar properties and weights, it was deemed necessary to simulate the manufacturing process to make the best educated choice for the concept selection. This will be covered in the next section.

Additive Manufacturing Process

To gain a full overview of the characteristics of the different driver concepts, printing simulations were performed using the Simufact software. This can save a lot of time and cost to create the final part. The part should print as close as possible to the intended design, minimising distortion and stress while keeping manufacturing time and costs low by minimising the required support structure and nesting. Overhangs are possible in LPBF, however, a steep overhang angle should be prevented. Smaller overhangs or no overhangs at all would be preferred. Bridges significantly reduce print quality when exceeding a few millimetres. Larger bridges are possible with the addition of supports; however, reducing bridges would make the model cheaper to print. Also, deviation of the print from the proposed design must be minimised. This must be done by having a minimum wall thickness of 1mm and reducing long and thin features.

2.1 Initial Simufact Simulations

To gain some insight into the printing process of the different driver concepts, an initial simulation process was set up. Below are the initial steps taken to simulate the laser powder bed fusion process of the drivers.

1. Import the part as a STEP-file
2. Centre the part on the base plate with 5mm height clearance
3. Apply material TI-AI6-V4
4. Use orientation assistant and apply wall thickness of 1mm and an overhang angle of 45 degrees
5. Generate supports
6. Apply support removal for all supports
7. Apply distortion compensation with 0.1
8. Generate 1mm uniform voxel mesh
9. Run simulation

2.2 Generative Design Simulations

The first simulation was performed on the generative design, using the process as described in Section 2.1. The model looks like a complicated part to print in its initial presentation, having big overhang angles and smaller components. The orientation assistant was used to find the optimal orientation for LPBF, minimising overhang angles and smaller structures, and keeping supports to a minimum. A 5 mm offset from the building plate was added for better removal. Build costs are calculated outside of Simufact. This led to a build orientation as seen in Figure 12.

When simulating the part initially, it was discovered that the printing process introduced some negative tolerances, this issue was solved by changing the acceptable distortion to 0.05. Another issue in the print regarded one of the mountings, where a lot of displacement occurred as can be seen in Figure 13.

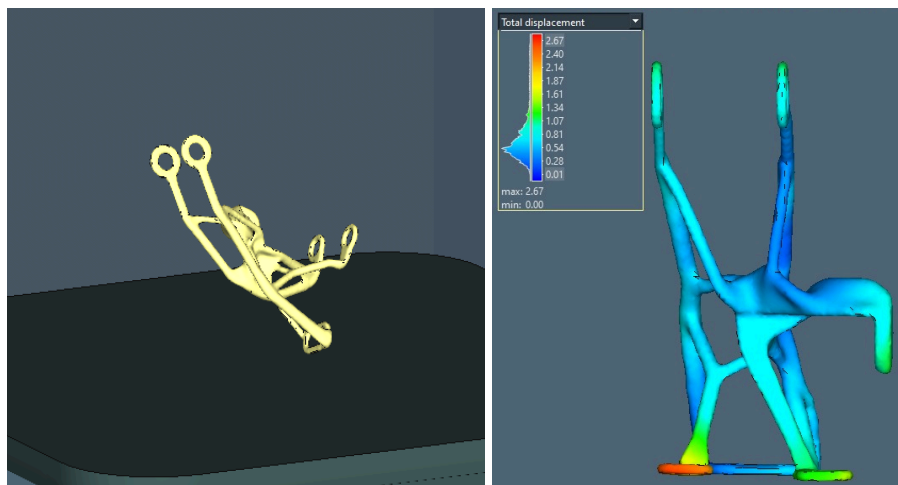


Figure 12: Orientation generative design simulation & Figure 13: Displacement at mounting in generative design.

To prevent this shape deviation during printing, a spar was added in an updated version of the generative design model. This spar links the two front mountings together. This resulted in a decrease of the maximum displacement from 2.67mm to 2.19mm and a lowered displacement curve, as can be seen in Figure 14.

To optimise the use of the printer and manufacture the driver in bigger batches, nesting was simulated. A distance of 5 mm between drivers and the building plate was to make sure all parts would come out well. A total of 16 parts could fit onto one building plate given the set preferences for printing.

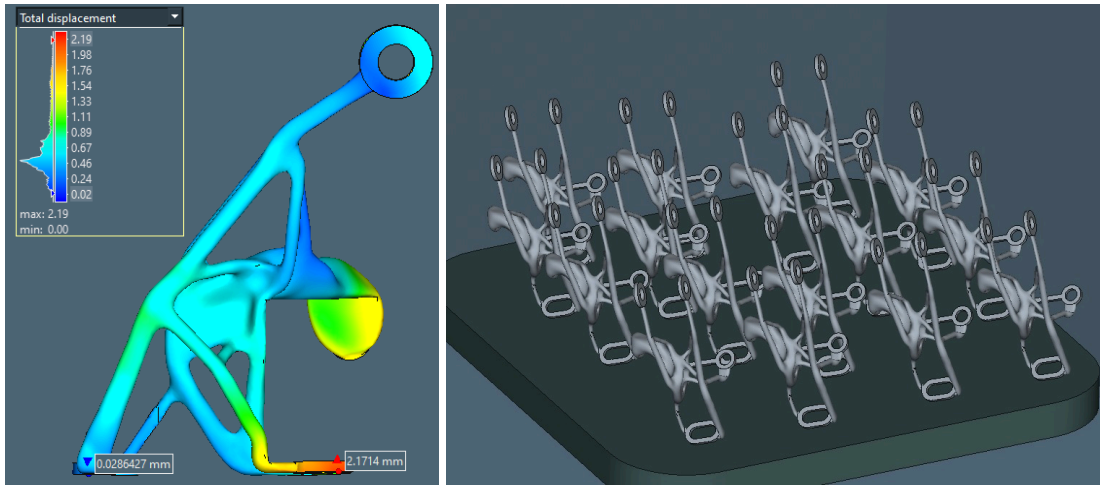


Figure 14: Updated generative design displacement & Figure 15: Nesting improved generative design.

2.3 Manual Design Simulations

The manual design looks quite similar from an LPBF printing side of view, just like the generative design with big overhangs and smaller components. The orientation process for the part was quite similar too, resulting in another angled design to minimise overhangs and greater angles.

The manual design showed a maximum displacement of 2.70 mm with a curve indicating a higher average displacement when tested for printing. Unlike the generative design, this part does not have one single place where displacement is very apparent, but rather a large spread over the sensor mounting and front mountings combined as can be seen in Figure 17.

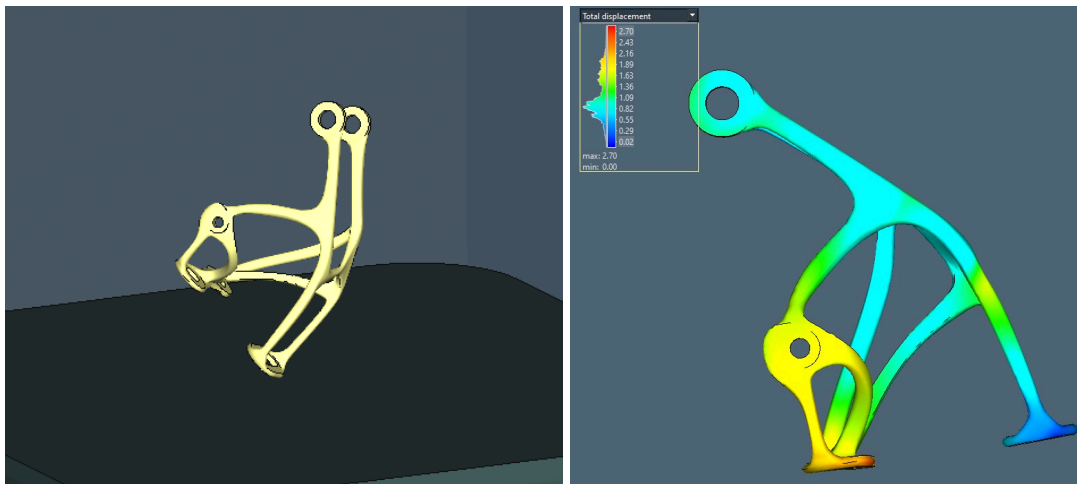


Figure 16: Orientation manual design simulation & Figure 17: Shape comparison manual driver concept

When simulating nesting the manually designed driver using the same setup as was used for the generative design, a total of 12 drivers would fit on the building plate.

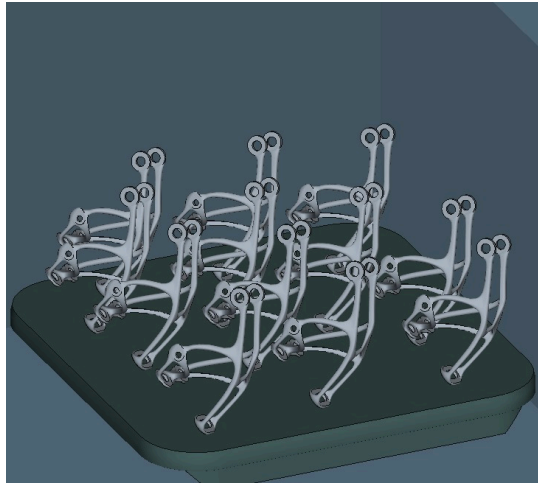


Figure 18: Nesting manually designed driver

Cost Analysis

Different factors determine and influence the cost of Additive Manufacturing parts, which range from material characteristics to process parameters, machine-related costs and post-processing requirements. However, it is possible to make an estimation, useful for decision-making and choosing one iteration over another. Cost per unit is calculated as a function of part volume and cost of material (expressed in €/cm³). Assuming a cost of 5,11 €/cm³ for L-PBF Titanium (5,68 - 10% (Vaneker et al. (2020))) the following estimations can be made:

Freeform iteration

-Cost per unit without supports

$$C(\text{per unit}) = \text{Volume} \times \text{Material Cost} = 22,83 \text{ cm}^3 \times 5,11 \text{ €/cm}^3 = 116,67\text{€}$$

-Cost per unit with supports

$$C(\text{per unit}) = \text{Volume with supports} \times \text{Material cost} = 55,52 \text{ cm}^3 \times 5,11 \text{ €/cm}^3 = 283,7 \text{ €}$$

-Cost per unit of post-processing that accounts for:

- Removal of supports in correspondence of non-critical surfaces (30€ for each surface) $30\text{€} \times 3 \text{ surfaces} = 90\text{€}$
- Removal of supports in correspondence of critical surfaces (90€ for each surface) $90\text{€} \times 9 \text{ surfaces} = 810\text{€}$
- Reworking of critical surfaces that show deformation or dimension misalignment of more than 0.1mm (100€ for each surface) $100\text{€} \times 4 \text{ surfaces} = 400\text{€}$

-Final cost per unit is estimated to be:

$$C = \text{Cost}(\text{per unit}) + \text{Cost of post processing}(\text{per unit}) = 1583,7\text{€}$$

Generative Design iteration

-Cost per unit without support

$$C(\text{per unit}) = \text{Volume} \times \text{Material Cost} = 22,49 \text{ cm}^3 \times 5,11 \text{ €/cm}^3 = 114,92\text{€}$$

-Cost per unit with supports

$$C(\text{per unit}) = \text{Volume with supports} \times \text{Material cost} = 46,06 \times 5,11 \text{ €/cm}^3 = 235,37 \text{ €}$$

-Cost of post-processing (per unit) that accounts for the aforementioned processes:

- $30\text{€} \times 4 \text{ surfaces} = 120\text{€}$
- $90\text{€} \times 10 \text{ surfaces} = 900\text{€}$
- $100\text{€} \times 6 \text{ surfaces} = 600\text{€}$

-Final cost per unit is estimated to be:

$$C = \text{Cost}(\text{per unit}) + \text{Cost of post processing}(\text{per unit}) = 1855,37\text{€}$$

In both cases, surfaces regarded as “critical” are the ones in correspondence of attachment points (sensor etc.) where displacement and/or dimensions out of tolerance can be more significant in assembly accuracy. “Non-critical” surfaces, instead, are all other areas where supports are touching or displacement of 0.1mm is negligible.

Chosen Design

The decision on which concept to proceed with is based on the properties of each driver, as can be seen in Table 2. The weight of the driver is an important consideration, along with the mechanical properties such as strength. In addition, the suitability of the model for LPBF printing is an important consideration.

Table 2: Comparison of Generative design an manual freeform concepts

	Generative design	Manual freeform
Weight	60.8 grams	102.83 grams
min. Safety Factor	1.5	1.69
Max. Stress	160 MPa	522.20 MPa
Max. displacement	1.2 mm	0.78 mm
LPBF max. displacement	2.19mm	2.70 mm
Nesting	16 parts per build	12 parts per build
Price per part	€1855,37	€1583,70

The generative design, although lighter and better printed, does not perform as well as the manually created part under load. Both designs are well within the required weight

reduction, so going for the somewhat heavier option should be acceptable and will save €271,67 in manufacturing per part which is 14,6% of the total manufacturing cost.

Discussion

Although the assignment aimed for a topology-optimised part, better results were found when using generative design and manual modelling. The final result shows a significant weight reduction to the original driver design while increasing its safety factor and consolidating multiple parts into one. However, the final design has a larger safety factor than required in the aircraft industry, meaning an even bigger reduction of weight should be possible within the margins. Simulating the LPBF process showed issues regarding displacement, which can not be removed but should be minimised. Although the curve of displacement and maximum displacement have been reduced, improvements could still be made. Finally, the LPBF process makes manufacturing quite expensive, especially the support removal and reworking of critical surfaces adds to this. Reducing critical surfaces or changing the design to meet only a 1.5 safety factor could make manufacturing somewhat cheaper.

Future directions and recommendations

The proposed model is an improvement on the original driver in terms of weight and part numbers. However, there are still some aspects that could benefit from additional optimization or require validation. The first aspect that could be improved is the design itself. The first thing that could be done is the refinement of the safety factors. A more detailed approach could reduce the areas where the safety factor is above 1.5. By removing material in these areas, the weight of the final model could be further reduced. The model could also be improved by introducing lattice or scaffolding structures, which could further decrease the weight of the driver.

The second aspect that could be improved is the manufacturing process. Currently, Simufact generates the build orientation and the supports. However, the orientation and support could be optimised. The removal of the supports of the critical surfaces is an expensive practice. The build orientation could reduce the amount of supports which are on the critical surfaces. Although this could increase the build time and build volume. Although this increases the costs, an analysis is required to determine whether this change would reduce the final costs of the driver. Besides the optimisation, the print method and supports also need to be validated. It needs to be tested whether this print causes cracks or creates a significant porosity.

Finally, aside from the validation of the print process, the mechanical properties of the driver also need to be validated. The current driver is based on a static analysis, but an airplane is a dynamic object. Therefore, a dynamic-, vibration- and fatigue analysis are required as future steps for the mechanical validation of the driver, and the suitability of the driver as an aircraft component.

Conclusion

Different methods for creating a new aircraft driver have been considered, including topology optimisation, generative design, a combination of the two and manual design based on the topology-optimised shape. Topology optimisation did not yield great results, although the material placement was optimised, too much material remained and disconnected when that material was removed. Next, similar constraints were applied to generative design while aiming for a 1.5 safety factor and weight reduction. This generative design was a well-connected part reducing the weight to 60 grams. Another generative design was created using a safety factor of 7, resulting in a bulkier model which in turn topology optimisation was applied on. This topology-optimised generative design however suffered from the same issues of the initial topology optimisation, as it applied large chunks of material for stiffness and removed connections first, and was limited by its design space which didn't allow material to be deposited on optimal spots. Finally, a driver model was manually created taking important characteristics from the results of the other processes into account. This process led to a model weighing 102.68 grams with a minimum safety factor of 1.69. The models were then put into a static simulation to evaluate their effectiveness and decide which one had the best properties.

Afterwards, the generative design and the manually created design were simulated for the laser-powder bed fusion additive manufacturing process. The generative design showed a large displacement at one of the mountings, which was countered by sparring it to the neighbouring mounting, resulting in a maximum displacement of 2.19 mm with a relatively lower displacement curve. The manually designed model did not have a single standout feature displacing, instead, the whole part showed a bigger displacement curve with a maximum of 2.70 mm.

When analysing costs, the generative design has fewer supports and less volume, but due to its greater amount of reworking and post-processing required it still manages to be a more expensive part to manufacture compared to the manually designed part.

The chosen design is the manually created design because it costs less to manufacture, can withstand more stress with a higher safety factor in a static test and doesn't exceed the intended weight reduction goals. Although it shows some more displacement when printing, the post-processing is still cheaper.

Sources

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